ADV: Logs and Exponentials (Adv), E1 Logs and Exponentials (Adv)

Log Calculus (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 73.5 minutes *(based on HSC allocation of 1.5 minutes approx. per mark)*

SUMMARY REVISION

- This subtopic is a database of *Log Calculus* questions that covers logarithmic differentiation and integration in the one location.
- It is important to note that this sub-topic is a collection of all questions within *Calculus C2 and C4* that involve logs, providing a convenient location for comprehensively revising log calculus.

HSC ANALYSIS - What to expect and common pitfalls

- *Log Calculus (1.8%)* has appeared in 7 exams in the last decade (most recently in 2020) and represents a great opportunity for high scoring.
- Log integration is more challenging than the simplistic log differentiation and is examined much more regularly.
- **More specifically, the integration of a fraction that results in a logarithm is a 2-3 mark question type that is regularly tested and deserves attention.**
- We note *2019 HSC 13c(ii)* surprisingly produced a sub-50% mean mark and should be reviewed.
- The new syllabus explicitly mentions the differentiation of $\log_a x$ and is covered in the database within SM-Bank questions.

Questions

1. Calculus, 2ADV C4 2018 HSC 5 MC

What is the derivative of $\sin(\ln x)$, where $x > 0$?

(A) $\cos\left(\dfrac{1}{x}\right)$

(B) $\cos(\ln x)$

(C) $\cos\left(\dfrac{\ln x}{x}\right)$

(D) $\dfrac{\cos(\ln x)}{x}$

---

2. Calculus, 2ADV C2 EQ-Bank 4 MC

If $f(x) = \log_2\left(x^{2x}\right)$, which expression is equal to $f'(x)$?

(A) $\dfrac{2}{x^{2x}\ln 2}$

(B) $\dfrac{2}{\ln 2} + 2\log_2 x$

(C) $\log_2 x + \dfrac{2}{\ln 2}$

(D) $\dfrac{2}{\ln 2} \times \log_2\left(x^{2x-1}\right)$

---

3. Calculus, 2ADV C4 2012 HSC 9 MC

What is the value of $\displaystyle\int_1^4 \frac{1}{3x}\,dx$?

(A) $\dfrac{1}{3}\ln 3$

(B) $\dfrac{1}{3}\ln 4$

(C) $\ln 9$

(D) $\ln 12$

---

4. Calculus, 2ADV C2 2008 HSC 2aii

Differentiate with respect to x:

$x^2 \log_e x$ *(2 marks)*

5. Calculus, 2ADV C2 SM-Bank 8

Let $y = (x+5)\log_e(x)$.

Find $\dfrac{dy}{dx}$ when $x = 5$. *(2 marks)*

6. Calculus, 2ADV C2 2011 HSC 1d

Differentiate $\ln(5x+2)$ with respect to x. *(2 marks)*

7. Calculus, 2ADV C2 2012 HSC 12ai

Differentiate with respect to x

$(x-1)\log_e x$ *(2 marks)*

8. Calculus, 2ADV C2 2015 HSC 11f

Differentiate $y = (x+4)\ln x$. *(2 marks)*

9. Calculus, 2ADV C4 2015 HSC 11h

Find $\displaystyle\int \frac{x}{x^2-3}\,dx$. *(2 marks)*

10. Calculus, 2ADV C2 2017 HSC 11d

Differentiate $x^3 \ln x$. *(2 marks)*

11. Calculus, 2ADV C2 EQ-Bank 1

Differentiate $\log_2 x^2$ with respect to x. *(2 marks)*

12. Calculus, 2ADV C4 2008 HSC 3b

i. Differentiate $\log_e(\cos x)$ with respect to x. *(2 marks)*

ii. Hence, or otherwise, evaluate $\displaystyle\int_0^{\frac{\pi}{4}} \tan x\,dx$. *(2 marks)*

13. Calculus, 2ADV C2 SM-Bank 6

Differentiate with respect to x:

$\log_e x^x$. *(2 marks)*

14. Calculus, 2ADV C4 2004 HSC 3bii

Find $\displaystyle\int \frac{x}{x^2-3}\,dx$. *(2 marks)*

15. Calculus, 2ADV C4 2006 HSC 2bii

Evaluate $\displaystyle\int_0^3 \frac{8x}{1+x^2}\,dx$. *(3 marks)*

16. Calculus, 2ADV C4 2020 HSC 17

Find $\displaystyle\int \frac{x}{4+x^2}\,dx$. *(2 marks)*

17. Calculus, 2ADV C4 SM-Bank 1

If $m = \displaystyle\int_1^3 \frac{2}{x}\,dx$, express e^m in its simplest form. *(2 marks)*

18. Calculus, 2ADV C4 2005 HSC 2ci

Find $\displaystyle\int \frac{6x^2}{x^3+1}\,dx$. *(2 marks)*

19. Calculus, 2ADV C4 2008 HSC 2ci

Find $\displaystyle\int \frac{dx}{x+5}$. *(1 mark)*

20. Calculus, 2ADV C4 2010 HSC 2dii

Find $\displaystyle\int \frac{x}{4+x^2}\,dx$. *(2 marks)*

21. Calculus, 2ADV C4 2011 HSC 4b

Evaluate $\displaystyle\int_e^{e^3} \frac{5}{x}\,dx$ *(2 marks)*

---

22. Calculus, 2ADV C4 2012 HSC 12b

Find $\displaystyle\int \frac{4x}{x^2+6}\,dx$. *(2 marks)*

---

23. Calculus, 2ADV C4 2013 HSC 11f

Evaluate $\displaystyle\int_0^1 \frac{x^2}{x^3+1}\,dx$ *(3 marks)*

---

24. Calculus, 2ADV C4 2019 HSC 13c

i. Differentiate $(\ln x)^2$. *(2 marks)*

ii. Hence, or otherwise, find $\displaystyle\int \frac{\ln x}{x}\,dx$. *(1 mark)*

Worked Solutions

1. Calculus, 2ADV C4 2018 HSC 5 MC

$$\begin{aligned} y &= \sin(\ln x) \\ \frac{dy}{dx} &= \cos(\ln x) \times \frac{d}{dx}(\ln x) \\ &= \cos(\ln x) \times \frac{1}{x} \\ &= \frac{\cos(\ln x)}{x} \end{aligned}$$

$\Rightarrow D$

---

2. Calculus, 2ADV C2 EQ-Bank 4 MC

$$\begin{aligned} f(x) &= \log_2\left(x^{2x}\right) \\ &= 2x\log_2 x \\ &= \frac{2x\ln x}{\ln 2} \end{aligned}$$

$$\begin{aligned} f'(x) &= \frac{1}{\ln 2}\left(2x \cdot \frac{1}{x} + 2\ln x\right) \\ &= \frac{2}{\ln 2} + \frac{2\ln x}{\ln 2} \\ &= \frac{2}{\ln 2} + 2\log_2 x \end{aligned}$$

$\Rightarrow B$

Worked Solutions

3. Calculus, 2ADV C4 2012 HSC 9 MC

$$\begin{aligned}&\int_1^4 \frac{1}{3x}\,dx\\ &= \frac{1}{3}[\ln x]_1^4\\ &= \frac{1}{3}[\ln 4 - \ln 1]\\ &= \frac{1}{3}\ln 4\end{aligned}$$

$\Rightarrow B$

TIP: Note that $\ln(1) = 0$, as $e^0 = 1$

4. Calculus, 2ADV C2 2008 HSC 2aii

$$\begin{aligned} y &= x^2 \log_e x\\ \frac{dy}{dx} &= x^2 \cdot \frac{1}{x} + 2x \cdot \log_e x\\ &= x + 2x\log_e x\end{aligned}$$

5. Calculus, 2ADV C2 SM-Bank 8

$$\begin{aligned}\frac{dy}{dx} &= 1 \times \log_e x + (x+5)\cdot\frac{1}{x}\\ &= \log_e x + \frac{x+5}{x}\end{aligned}$$

$$\therefore \frac{dy}{dx}\,|_{\,x=5} = \log_e 5 + 2$$

6. Calculus, 2ADV C2 2011 HSC 1d

$$\begin{aligned} y &= \ln(5x+2)\\ \frac{dy}{dx} &= \frac{5}{5x+2}\end{aligned}$$

7. Calculus, 2ADV C2 2012 HSC 12ai

$$\begin{aligned} y &= (x-1)\log_e x\\ \frac{dy}{dx} &= 1(\log_e x) + (x-1)\frac{1}{x}\\ &= \log_e x + 1 - \frac{1}{x}\end{aligned}$$

8. Calculus, 2ADV C2 2015 HSC 11f

$y = (x+4)\ln x$

Using the product rule

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x+4)\cdot\ln x + (x+4)\frac{d}{dx}\ln x\\ &= \ln x + (x+4)\frac{1}{x}\\ &= \ln x + \frac{4}{x} + 1\end{aligned}$$

9. Calculus, 2ADV C4 2015 HSC 11h

$$\begin{aligned}&\int \frac{x}{x^2-3}\,dx\\ &= \frac{1}{2}\int \frac{2x}{x^2-3}\,dx\\ &= \frac{1}{2}\ln\left(x^2-3\right) + C\end{aligned}$$

10. Calculus, 2ADV C2 2017 HSC 11d

$y = x^3 \ln x$

Using the product rule:

$$\begin{aligned}\frac{dy}{dx} &= 3x^2\cdot\ln x + x^3\cdot\frac{1}{x}\\ &= x^2(3\ln x + 1)\end{aligned}$$

11. Calculus, 2ADV C2 EQ-Bank 1

TIP: The new Advanced reference sheet can be used here!

$$\begin{aligned} y &= \log_2 x^2 \\ \frac{dy}{dx} &= \frac{d}{dx}\left(\frac{\ln x^2}{\ln 2}\right) \\ &= \frac{1}{\ln 2} \cdot \frac{d}{dx}(\ln x^2) \\ &= \frac{1}{\ln 2} \cdot \frac{2x}{x^2} \\ &= \frac{2}{x \ln 2} \end{aligned}$$

12. Calculus, 2ADV C4 2008 HSC 3b

i.

$$\begin{aligned} y &= \log_e(\cos x) \\ \frac{dy}{dx} &= \frac{-\sin x}{\cos x} \\ &= -\tan x \end{aligned}$$

ii. $\displaystyle\int_0^{\frac{\pi}{4}} \tan x \, dx$

$$\begin{aligned} &= -\left[\log_e(\cos x)\right]_0^{\frac{\pi}{4}} \\ &= -\left[\log_e\left(\cos\left(\frac{\pi}{4}\right)\right) - \log_e(\cos 0)\right] \\ &= -\left[\log_e\left(\frac{1}{\sqrt{2}}\right) - \log_e 1\right] \\ &= -\left[\log_e\left(\frac{1}{\sqrt{2}}\right) - 0\right] \\ &= -\log_e\left(\frac{1}{\sqrt{2}}\right) \\ &= 0.346\ldots \\ &= 0.35 \ \ (2 \text{ d.p.}) \end{aligned}$$

13. Calculus, 2ADV C2 SM-Bank 6

$$\begin{aligned} y &= \log_e x^x \\ &= x \log_e x \\ \frac{dy}{dx} &= x \cdot \frac{1}{x} + \log_e x \\ &= 1 + \log_e x \end{aligned}$$

14. Calculus, 2ADV C4 2004 HSC 3bii

$$\begin{aligned} \int \frac{x}{x^2-3}\, dx &= \frac{1}{2}\int \frac{2x}{x^2-3}\, dx \\ &= \frac{1}{2}\ln(x^2-3) + C \end{aligned}$$

15. Calculus, 2ADV C4 2006 HSC 2bii

$$\begin{aligned} &\int_0^3 \frac{8x}{1+x^2}\, dx \\ &= 4\int_0^3 \frac{2x}{1+x^2}\, dx \\ &= 4\left[\log_e(1+x^2)\right]_0^3 \\ &= 4[\log_e(1+9) - \log_e(1+0)] \\ &= 4[\log_e 10 - \log_e 1] \\ &= 4\log_e 10 \end{aligned}$$

16. Calculus, 2ADV C4 2020 HSC 17

$$\begin{aligned} \int \frac{x}{4+x^2}\, dx &= \frac{1}{2}\int \frac{2x}{4+x^2}\, dx \\ &= \frac{1}{2}\log_e(4+x^2) + c \end{aligned}$$

17. Calculus, 2ADV C4 SM-Bank 1

$$\int_1^3 \frac{2}{x}\,dx = [2\log_e x]_1^3$$

$$m = 2\log_e 3 - 2\log_e 1$$
$$= 2\log_e 3$$
$$\therefore e^m = e^{2\log_e 3}$$
$$= e^{\log_e 9}$$
$$= 9$$

---

18. Calculus, 2ADV C4 2005 HSC 2ci

$$\int \frac{6x^2}{x^3+1}\,dx = 2\int \frac{3x^2}{x^3+1}\,dx$$
$$= 2\ln\left(x^3+1\right) + C$$

---

19. Calculus, 2ADV C4 2008 HSC 2ci

$$\int \frac{dx}{x+5}$$
$$= \ln(x+5) + C$$

---

20. Calculus, 2ADV C4 2010 HSC 2dii

$$\int \frac{x}{4+x^2}\,dx$$
$$= \frac{1}{2}\int \frac{2x}{4+x^2}\,dx$$
$$= \frac{1}{2}\ln\left(4+x^2\right) + C$$

IMPORTANT: Minimise errors by adjusting the integral to fit the form $\dfrac{f'(x)}{f(x)}$ *before* integrating.

---

21. Calculus, 2ADV C4 2011 HSC 4b

$$\int_e^{e^3} \frac{5}{x}\,dx$$
$$= 5\int_e^{e^3} \frac{1}{x}\,dx$$
$$= 5[\ln x]_e^{e^3}$$
$$= 5\left(\ln e^3 - \ln e\right)$$
$$= 5(3-1)$$
$$= 10$$

MARKER'S COMMENT: Most common error was $\ln(5x)$. Minimize errors by getting the integral in the form of $\dfrac{f'(x)}{f(x)}$ *before* integrating.

---

22. Calculus, 2ADV C4 2012 HSC 12b

$$\int \frac{4x}{x^2+6}\,dx$$
$$= 2\int \frac{2x}{x^2+6}\,dx$$
$$= 2\ln\left(x^2+6\right) + C$$

---

23. Calculus, 2ADV C4 2013 HSC 11f

$$\int_0^1 \frac{x^2}{x^3+1}dx$$
$$= \frac{1}{3}\int_0^1 \frac{3x^2}{x^3+1}dx$$
$$= \frac{1}{3}\left[\ln\left(x^3+1\right)\right]_0^1$$
$$= \frac{1}{3}(\ln 2 - \ln 1)$$
$$= \frac{1}{3}\ln 2$$

TIP: Note that $\ln(1) = 0$, because $e^0 = 1$

---

24. Calculus, 2ADV C4 2019 HSC 13c

i. $y = (\ln x)^2$

$$\frac{dy}{dx} = 2 \cdot \frac{1}{x} \cdot \ln x$$

$$= \frac{2\ln x}{x}$$

ii. $\displaystyle\int \frac{\ln x}{x}\, dx = \frac{1}{2}\int \frac{2\ln x}{x} dx$

$$= \frac{1}{2}(\ln x)^2 + C$$

♦ *Mean mark part (ii) 49%.*